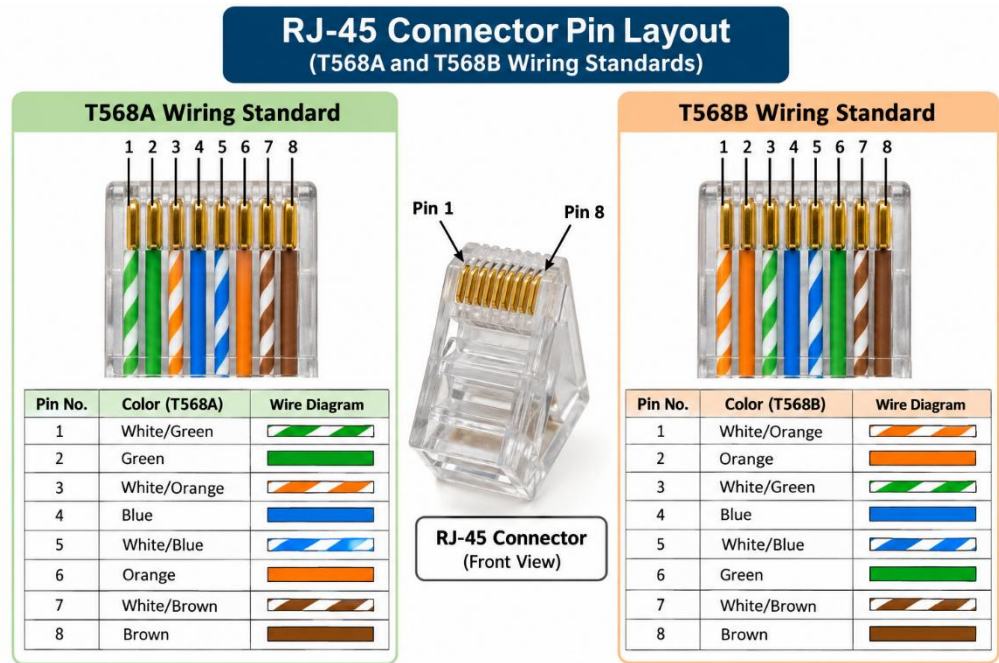


Ethernet standard and specifications

1. Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.



2. Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Ethernet Standard	Speed	Cable Type	Maximum Cable Length	Connector Type
10BASE-T	10 Mbps	Cat3 or better UTP	100 meters	RJ-45
100BASE-TX	100 Mbps	Cat5 or better UTP/STP	100 meters	RJ-45
1000BASE-T	1 Gbps	Cat5e or better UTP	100 meters	RJ-45

3. Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.

Direct Laptop-to-Laptop Ethernet Connection

To connect two laptops directly using Ethernet, I would use a **crossover cable**. A crossover cable is designed to connect two similar devices, such as two laptops, without using a switch or router. It works by **swapping the transmit (TX) and receive (RX) wire pairs**, so the

transmit pins of one laptop connect to the receive pins of the other laptop. This allows both laptops to communicate and exchange data directly.

Cable used: Crossover Ethernet Cable

Reason: It crosses the TX and RX pairs between the two laptops, enabling direct communication.

4. List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

Real-World Devices Using Ethernet Connections

No.	Device	Ethernet Standard	Typical Use
1.	Wi-Fi Router	1000BASE-T	Connects the router to a computer or network switch using Ethernet.
2.	Desktop Computer	1000BASE-T	Provides a wired connection to the internet or local network.
3.	Network Printer	100BASE-TX / 1000BASE-T	Connects the printer to the local network so multiple users can print.

Conclusion: These devices commonly use **RJ-45 Ethernet connections**, with **1000BASE-T (Gigabit Ethernet)** being common in modern networking equipment.